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Olive Spot Disease Detection and Classification using Analysis of Leaf Image Textures

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Abstract

The olive tree is a highly beneficial fruit tree with the earliest known history of its plantation going back to 6000 years. The production of olive oil is facing a significant threat nowadays due to climate change and the spread of diseases. In this paper, olive disease analysis and classification using image processing techniques are done. Using image texture analysis of olive plant leaves, correlation in between the signatures of Neofabrea leaf spot disease and Peacock leaf spot diseases with some texture features is identified. Neofabrea leaf spot disease manifests itself as distinguishable spots, prominently circular but is quite similar to the peacock leaf spot disease. In the proposed methodology, the infected area is isolated using the histogram thresholding and k-means segmentation. Out of the two methods, k-mean segmentation offered higher accuracy. In the next step, texture analysis is applied using first to fourth order moments, which in turn helped us to identify the relationship of infection with one or more texture features. A strong correlation between infection area and texture features such as energy, homogeneity, entropy is identified, which also helped in classifying the two similar diseases.

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1. Introduction

The olive, also known as *Olea Europaea* is the species belonging to genus *Olea*. The olive tree is expected to live around 400-600 years. Historically, olive trees were found in the eastern Mediterranean for the first time around 7000 years ago. Olive fruits are mainly used to produce olive oil, which has been shown to have excellent properties

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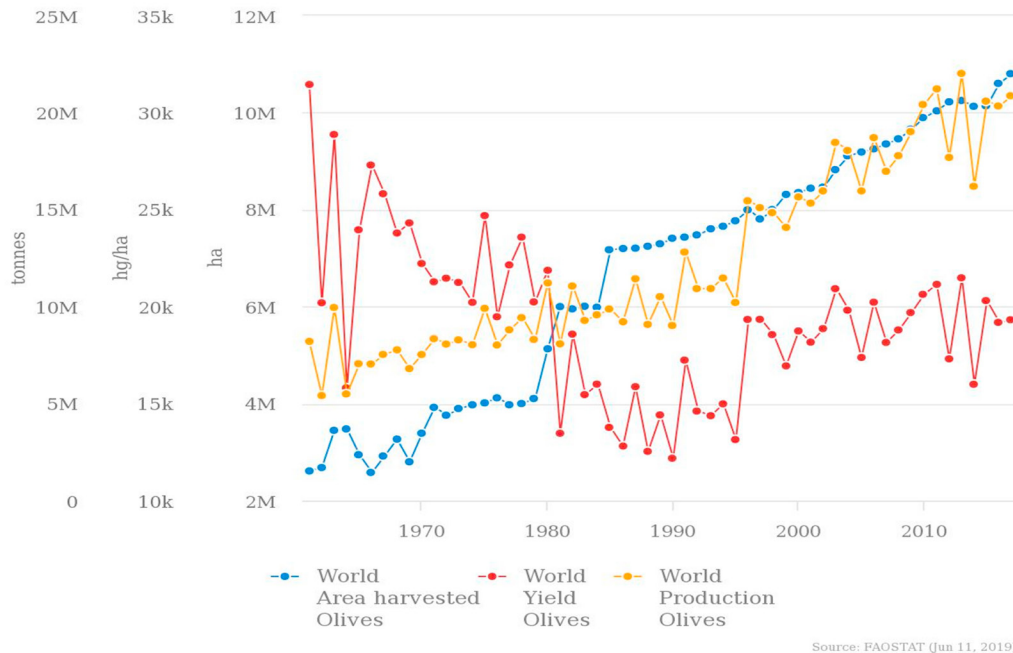


Fig. 1. Olive Yield Throughout Years[8]

Importance of Olive trees can be understood from its widespread growth and thus found in about 64 countries. Olive production and its harvesting have increased continuously from 1965 to 2017(Fig. 1). In the year 2017, olive production was around 20 million tonnes and the area harvested is more than 10 million hectares. However, to everybody's concern, the olive yield is not increasing consistently. It's yield in the year 1961 was 31453 hg/ha(hectograms (100 grams) per hectare), whereas, in the year 1990, it was down to 12179 hg/ha, and followed by that in 2017 to 19319 hg/ha(Fig. 1). The yield has not been consistent because of various reasons like unfavourable climate changes, disease infestation, pathogen attacks, to name a few.

Olive oil production in significant nations sees a steep decline. Like in Spain, it is the biggest producer the contribution has declined to almost 44% of the total olive production[18] compared to its earlier figures. In a recent study [18] it is estimated that nearly 20000 olive farms will vanish in the coming decade,. In recent times in Italy, olive production is nearly 57 % compared to the previous year[9]. With the spread of *Xylella fastidiosa* a specific type plant bacteria, significant threats are there on olive trees[1], an olive growing region in Italy has lost nearly 11 million olive trees, and still have no cure till date[3].

Olive is affected by many types of pathogens and deficiencies. Some of the olive diseases are bacteria wilt, olive knot, bacterial spot, bacterial leaf blight, angular leaf spot, verticillium wilt, olive leaf spot, leaf mould, olive fruit fly, olive bark beetle, olive borer, olive moth, nitrogen, calcium, magnesium, zinc deficiency and more.

Neofabrea and peacock leaf spot diseases are spot diseases which appear visually on the host plant leaves; In the case of Peacock leaf spot disease, young leaves are more vulnerable to the fungus than the older ones.

From Fig. 1 it is clear that the olive yield has decreased with respect to the area harvested for olive production. There is a lot of advancement in the field of plant disease detection using image processing techniques. But since disease is a very varying term the distinguishing factors will always vary and be from a vast super-set. So to get a general and working for all type of disease algorithm will always be a challenge, so in this work we are trying to identify on of the pieces of bigger puzzle. This identification of the correlation between texture features and spot disease will help solve classification of the diseases so that the remedy can be proper and quick for the specific identified disease.

1.1. Texture

The texture is the distinguishing factors in identifying an object or pattern of interest in an image. Texture can be smooth, silky, rough, coarse, bumpy or more in terms of touch. In terms of an image, texture can be a pattern of highs and lows on intensities at a localised area of an image or ridges and peaks in an intensity graph or its histogram. Texture in terms of the image depends on the scale of the window, which moves over the image and the direction it moves. Harlick et al.[10] were the first one to derive statistical texture features of an image. Harlick et al. devised 14 different textures features based on a Grey level co-occurrence matrix(GLCM). This GLCM matrix is based on the direction, and the distance and the features depend on them. Other people have derived other statistical features as well from this matrix. Some of the texture features we have used in this work are explained below.

Energy represents how many repetitive pixels are appearing in the image. So as the infection in the leaf is increasing the repetitive healthy pixels will decrease and hence homogeneity will decrease, and so the energy will decrease as well. Its range is [0-1]. The equation can be seen in Eq.1.

$$\sum_{x,y=0}^{N-1} (T_{xy})^2 \quad (1)$$

Entropy:Textural uniformity is measured using entropy value. As the complexity increases in the image, the value of entropy also increases. As the infection in the image increases the entropy value also increases. There is an inverse correlation between entropy and energy. Equation can be seen in Eq.2

$$\sum_{x,y=0}^{N-1} -\ln(T_{xy}) T_{xy} \quad (2)$$

Contrast

$$\sum_{x,y=0}^{N-1} T_{xy}(x-y)^2 \quad (3)$$

Homogeneity will decrease as the infection will increase. More the elements are closer in the GLCM matrix more the homogeneity value will be. Contrast values are inversely proportional to homogeneity. Equation for homogeneity can be seen in Eq.3.

$$\sum_{x,y=0}^{N-1} \frac{T_{xy}}{1 + (x-y)^2} \quad (4)$$

Correlation:The correlation texture feature represents a Linear relationship between the neighbouring pixels. If the infection is spread in one spot, then the correlation will be more, and so one can isolate the spread of disease and the localisation using the correlating feature value. The equation may be seen in Eq.5.

$$\sum_{x,y=0}^{N-1} T_{xy} \frac{(x-\mu)(y-\mu)}{\sigma^2} \quad (5)$$

Contrast:The intensity variation w.r.t. the neighbourhood pixel is called the contrast. Contrast also represents the spatial frequency of the image. The equation can be seen in Eq.6.

$$\sum_{n=0}^{N_g-1} n^2 \left(\sum_{x=1}^{N_g} \sum_{y=1}^{N_g} T(x,y) \right) \quad (6)$$

where:

$x, y = x^{th}$ row element of the matrix and y^{th} column element of the matrix

T_{xy} = Element x,y of the normalized symmetrical GLCM

N = Number of gray levels in the image

μ = the GLCM mean, calculated as:

$$\mu = \sum_{x,y=0}^{N-1} iT_{xy} \quad (7)$$

σ^2 = the variance of the intensities of all pixels in the GLCM, calculated as:

$$\sigma^2 = \sum_{x,y=0}^{N-1} T_{xy}(x - \mu)^2 \quad (8)$$

1.2. Literature Review

Haralick et al.[10] introduced 14 texture features to distinguish objects and region of interests in an image. Texture can be coarse, smooth, bumpy, silky as described by touch, but also can be incorporated in the image. Harlick also talked about the directionality of the pixel neighbourhood, i.e. formation of grey level co-occurrence matrix in all four directions(0° , 45° , 90° and 135°). Textures such as energy, angular second moment, entropy, sum of squares, sum variance, inverse difference moment, difference entropy and more. For determination of textures Connors et al.[6] compared four different texture determination algorithms and found out that Spatial Gray Level Dependence Method(SGLDM)performed much better than the rest of them since it included the class of Markov textures. Chen et al.[5]developed a technique, namely texture spectrum approach. In this technique, they determined the global texture of the image by texture spectrum, and for neighbourhood pixels, Chen et al. used the texture units. Chen et al. determined the values in all eight directions which was very much useful when tested on broadtz dataset[4]. Stachowicz et al.[16] used colour as a feature for segmentation and classification. A feature vector is created for each foreground, which was extracted from the original image; then it was compared with the database features. Stachowicz et al. achieved 99% accuracy in the best case scenario. Chen et al. [5] proposed a texture description method using graph containing texture image function. Proposed Statistical Landscape Features (SLF) achieved 94.7% of classification rate on the Brodatz dataset[4].Tivive et al.[17] presented a technique based on a convolutional neural network which extracted and classified the features in the same architecture. The proposed CoNN, when used images post-processed with median filtering performed much better than the Gabor filters, wavelets, quadratic mirror filters (QMF) and co-occurrence matrix methods.

Latest work in the detection of the plant diseases using image processing techniques is summarised here. Bai et al.[2] worked the leaf spot disease caused by bacteria which affects the cucumber plant using Fuzzy logic technique, Bai et al. used around 129 images for testing and training. Zhang et al.[20] identified the downy mildew, bacterial angular scab, gray mold, anthracnose and powdery mildew caused by bacteria and fungus on cucumber plant using sparse representation technique; Zhang et al. worked in 420 images in their dataset. Hassanein et al.[11] detected powdery mildew and early blight caused by fungus on tomato plant using moth flame optimisation; Hassanein et al. tested their result on UCI data repository. Dechant et al.[7] worked on Maize plant and detected Northern leaf blight fungus using deep learning techniques on a dataset of 1,834 images. Piret et al. Momin et al. & Maharlooei et al. [15, 14, 13] identified soybean diseases and detected defected beans, split beans and pods/stems, soybean aphids, mildew, rust, tan and rust R8. Johannes et al. and Wang et al. [12, 19] worked on wheat crop and detected rust, septoria and tan spots leaf nitrogen concentration using candidate hotspot detection and Multiple kernel SVM techniques.

The paper is further organised as Section 2 explains the methodology used to obtain the results in work, Section 3 includes the results, such as graphs charts and screenshots, obtained during this work, Section 4 concludes the paper with the summary of the work and the results obtained. In Section 5 future work which may be carried out with the devised methodology and Section 6 includes the acknowledgement.

2. Methodology

In this paper a specific technique is identifies which is used in analysing and classifying the spot diseases affecting the Olive plant. Correlation among textures features and spot diseases has been identified. Testing and detection of

the diseases has been carried out on the collection of olive leaf images which are affected by *Neofabrea* fungus and *Spilocaea oleaginea* fungus. Both the diseases exhibit similar infection patterns on the leaves. In case of Peacock spot the spot is visibly circular but sometimes irregular as well, for leaves affected by *neofabrea*, the infection pattern is spread in one spot and similar in colour to peacock spot. Healthy olive leaf images were collected from the local olive farm and infected images from the online sources. Eleven images of the peacock spot diseased leaves are collected and 12 images of the *neofabrea* leaf spot diseased leaf images.

The program application is developed in Matlab 2019a to isolate the ROI and for the texture analysis using GLCM matrix. Local images were collected using Nikon D5200 camera along with 18-55mm and 50mm prime lens and infected images are collected from online resources. For isolation of the infected area from the diseased leaf, two techniques are identified in this work, those are, k-means segmentation and histogram thresholding. As in the Fig.2 the ROI methodology part is explained.

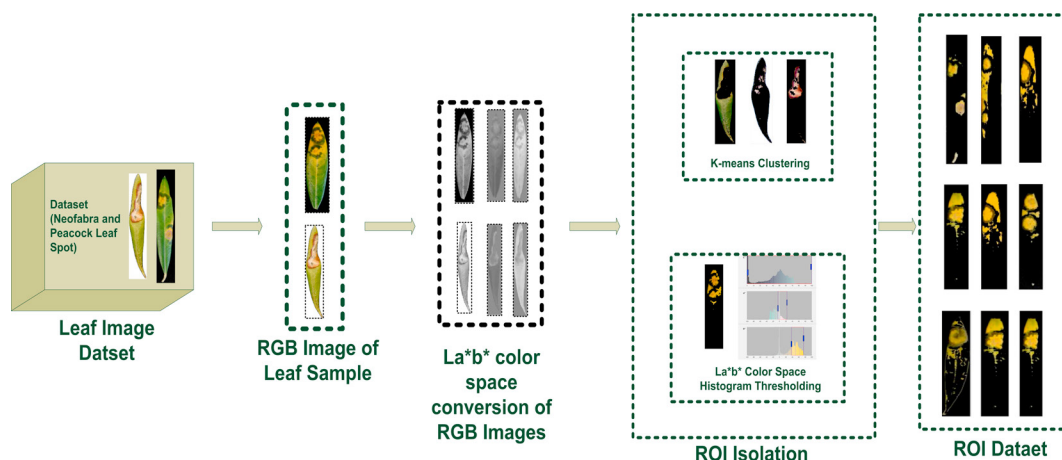


Fig. 2. Block diagram of the first stage of methodology

Infection on the leaf caused by both the diseases was quite similar visibly. In case of the *neofabrea* leafspot, the infection spread in nearly circular, the colour of the leaf changes from green to brown, reddish brown and sometimes almost fading brown Fig.4. The necrotic area is darkest on the outside and fades as the infection reaches the core. In case of the peacock spot the infection area is more prominently circular, also the colour of the infected area in the dark green outer layer then dark brown/black spots and then finally the infected area. The infected area of both diseases is quite similar to each other, so it is hard to distinguish between the two.

For the isolation of the infected area(ROI) in the diseased leaf, two major techniques were used and tested, those are k-means ROI segmentation and the thresholding on the La^*b^* colour space histogram values. After testing and verifying the result visibly, from the experimentation it was found out that k-means segmentation was able to isolate the ROI properly much effectively. In the Fig.5 as we can see the different results obtained by k-means segmentation on both sets of images affected by *neofabrea* and peacock spot disease, respectively.

For texture analysis, texture features as introduced firstly by Harlick et al.[10] were calculated. Texture features were based on the first to fourth moments on the Grey Level Co-Occurrence Matrix. Many features, as mentioned in the Introduction section, were calculated for the isolated infected area of both the set of diseased images. The correlation values were found out for each feature value with the infected area percentage as seen in the Fig.6.

3. Results

The ROI isolation results obtained using the k-mean segmentation based on three cluster colours can be seen the Fig.5. As one can see from Fig.5 infection area on leaves affected by *Neofabrea* leaf spot is isolated correctly, but in some cases, this technique struggles with the infection area on leaves affected by peacock spot. The dark green colour in the outer layer of the infection forces the algorithm to give bad results.

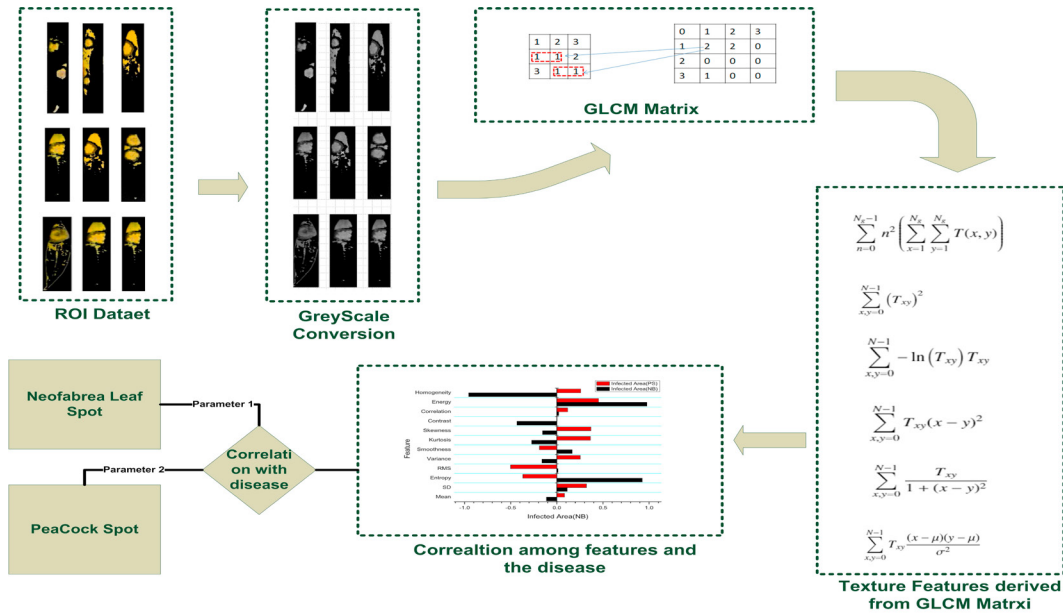


Fig. 3. Block diagram of the second stage of methodology

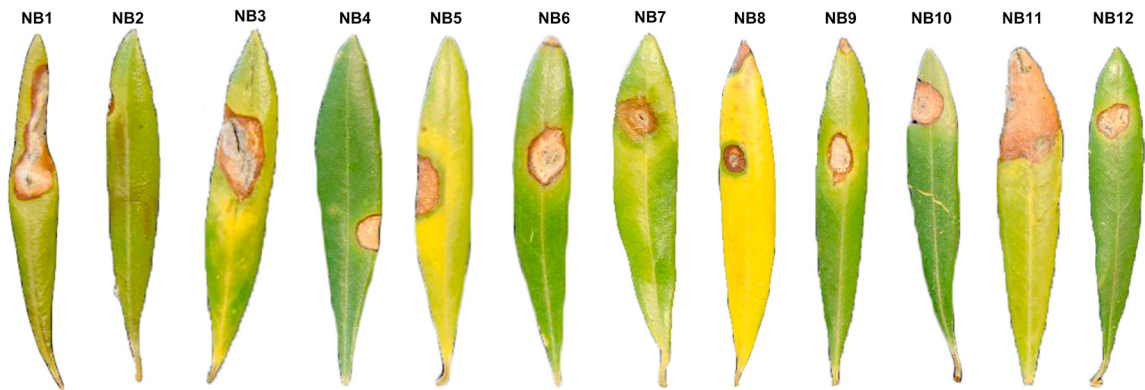


Fig. 4. Olive leaves affected by Neofabrea Leaf Spot

As looking in the Fig.6 energy and entropy has very high correlation values w.r.t. The infection percentage area for the leaf samples affected by neofabrea leaf spot. Also when we plot the energy and entropy values for each neofabrea sample; Fig.8 & Fig.7 the definition of the both features. As the infection area is increasing the uniformity is decreasing in the image, and so the energy values should decrease according to this definition, as one can see in the Fig.8 for each leaf sample the infection area can be seen and the corresponding energy values as well. The graph is plotted from the sorted column according to the infection area percentage w.r.t to each leaf sample affected by neofabrea leaf spot disease. As it clear from the slope of the graph, the definition of the energy holds. Similarly, for the entropy feature values, the randomness is increasing in the images which are more.

4. Conclusion

In this work, a very reliable method of identifying certain leaf spot type diseases occurring in olive trees is identified. Specific parameters and their values derived from the leaf image textures of the spots due to such disease also help in distinguishing the disease type as well of the same category. A high correlation with neofabrea leaf spot dis-

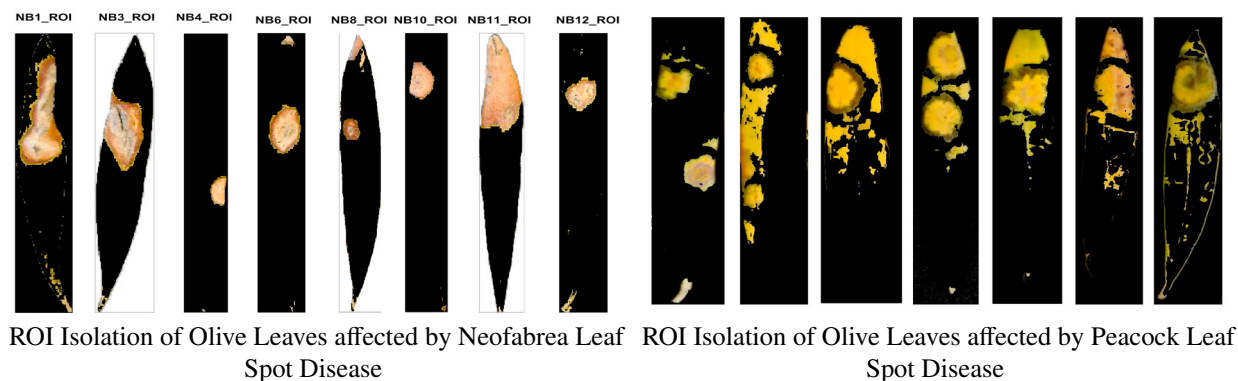


Fig. 5. ROI Isolation of Diseased Olive Leaves

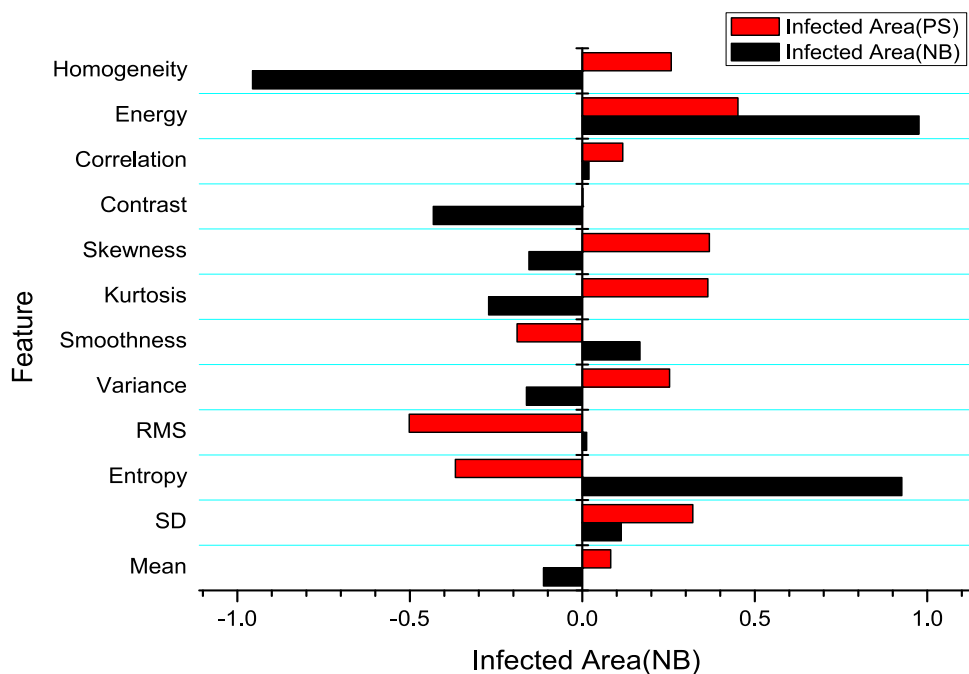


Fig. 6. Correlation values of each texture feature w.r.t. infected area of neofabrea and peacock leaf spot diseased leaf images

ease has been found of energy($r=0.97$) and entropy($r=0.92$) properties of spot texture. These parameters also help in classifying the disease type and thus distinguishing peacock leaf spot disease from neofabrea which in appearance is pretty similar otherwise, as the correlation values for the energy and entropy for peacock spot are $r=0.36$ and $r=0.45$. Further, ROI isolation was done using two techniques, namely, k-means segmentation and thresholding the La^*b^* colour space histogram values. The images of diseased leaves were collected from various sources as well as from local farms and thus offered a rich database to derive reliable predictions.

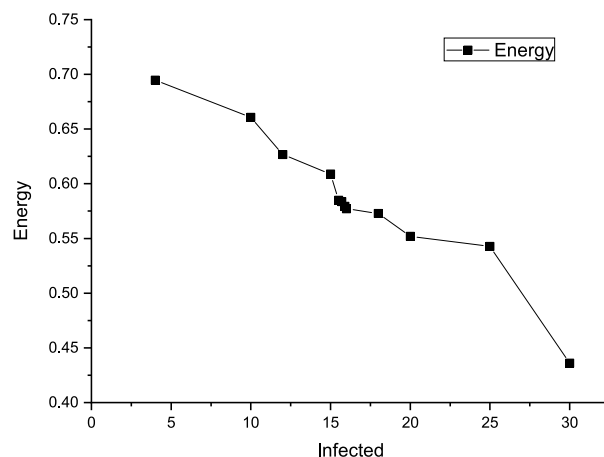
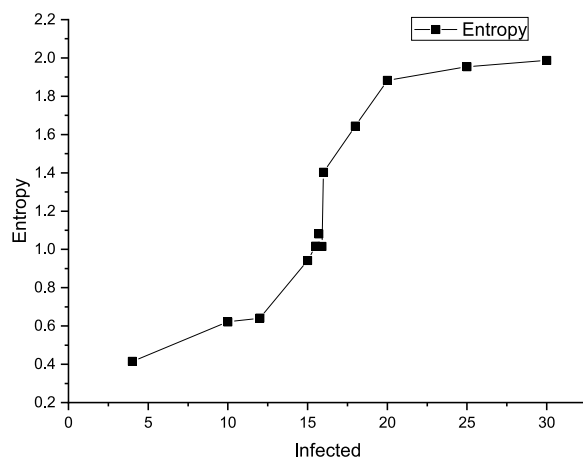


Fig. 7. Entropy values w.r.t. Infection Area for Neofabrea Spot Images

Fig. 8. Energy values w.r.t. Infection Area for Peacock Spot Images

5. Future Work

The work can be extended to other plants to predict and classify similar diseases with the above methodology. Detection of other diseases shall increase the confidence in the methods developed and increase its applicability. The study for olive tree can be further enhanced by including more data in varying conditions and climates to make the methodology more robust.

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